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ARTICLE

The Perspectival Realist features of Ernst Mach's critical epistemology

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Abstract
This paper has a twofold aim. On the one hand, it explores the extent to which Mach was inspired by Kant's approach to philosophical inquiry and tried to further elaborate it through his historico-critical method for enlightening scientific knowledge claims. On the other hand, it argues that the focus on the situated character of these claims that is implied in Mach's epistemology makes it possible to compare his view to recent attempts to defend a perspectival realist account of scientific knowledge, thus revealing the relevance of Mach's own approach as a methodology in the philosophy of science.

Keywords Epistemology · Scientific perspectivism · Scientific realism · Instrumentalism · Pragmatism · Economy of Thought

It is well known that Ernst Mach tried to avoid being called a philosopher–at least in the sense this term had in his time. His explicit aim was to contribute to a renewal of scientific inquiry that would obliterate metaphysical concepts of all kinds from science. However, it is uncontroversial that Mach's contribution to his discipline had–and continues to have–philosophical relevance, insofar as it is the expression of a proper philosophical approach to scientific knowledge as we currently conceive of it. Mach's epistemology, or "philosophy of science", is in fact focused on the methodology, the value, and the actual meaning of scientific knowledge claims, and both the way in which he pursued his aims and the outcomes of his work deeply inspired further reflection on these issues. It should come as no surprise, then, that Mach's view contains ideas that are still defended by contemporary philosophers of science, although within a different framework and, consequently, with a

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different scope and meaning.[1] In a modest way, it can be said that certain fundamental tenets of Mach's epistemology anticipated features that pertain to currently debated positions, in such a way that it may be possible to revitalize the former and put them into dialogue with contemporary viewpoints on science. This does not mean that we should isolate Mach from his historical and philosophical context, or that we are allowed to project onto him ideas that have been developed only recently. In both cases, the result would be an inconsistent and unfounded retrospective reading. Rather, I would like to argue that Mach's relevance to the history of philosophy (especially the philosophy of science) can be assessed by looking at the role that his most important ideas can still play in the current debate. That is to say, I believe that the "fruits" of Mach's original approach to scientific knowledge can still be encountered in current philosophy of science.[2]

In this paper, I will try to make the case for this idea by taking into account the perspectival realist view defended by Ronald Giere, Michela Massimi, and Hasok Chang. The first section will be devoted to the principles of Mach's historico-critical approach to scientific knowledge, which can be interpreted as an attempt to elaborate Kant's philosophical methodology in a new way and to make use of it to rid scientific inquiries of metaphysical obscurities. This study has a twofold aim: on the one hand, to provide further considerations, in addition to those examined in the existing literature, with which to assess the originality of Mach's methodological attitude toward scientific knowledge; on the other hand–and perhaps most importantly–to explore the basic tenets of Mach's epistemology, focusing especially on his interest in the value and meaning of scientific knowledge. Furthermore, I will argue that Mach's assessment of scientific concepts, theories, and laws and his observations on the relationship between theories and facts constitute a relevant contribution to the debate on scientific realism that is worthy of consideration. In fact, in the second section, I will argue that there is more than one feature of Mach's epistemological view that may be consistent with the way perspectival truth has been assessed in recent discussions, especially if one considers Mach's focus on the situatedness of scientific knowledge, both within its historical and cultural context and within the individual viewpoint determined by the interests of any researcher working in a well-defined field (e.g. physics vs. psychology).

[1] Erwin Hiebert (1970, 194), e.g., remarked that "the specific questions Mach raised about science were essentially of a philosophical nature. Although he was never at ease with 'philosophy' he was not at all averse to the examination of philosophical questions provided that such activity implied nothing more than 'scientific epistemology', i.e., *erkenntnistheoretische Untersuchungen*". Similarly, Richard von Mises (1987, 186) observed that Mach in fact "engaged in *philosophical activities*, at least if we regard it as the task of a philosophy to *clarify* statements made in daily life or in science, to analyse their *sense*, and to investigate their *connections*". He further maintained that "there is hardly a physicist today whose whole half-conscious, half-unconscious attitude towards the fundamental questions of his science has not been determined by Mach's questioning of dogmas and his concrete elucidations of the fundamental concepts of physics" (1987, 189).

[2] Max Plank concluded his attack against Mach's epistemology with the following sentence: "In the serene trust in the power of the Word, which for over nineteen hundred years has taught us the ultimate indubitable sign of how to distinguish true from false prophets: *By their fruits shall ye know them!*" (Blackmore 1992, 132).

1 A renewed philosophical method for science

Mach's relationship with the Western philosophical tradition and the role he played in its historical development can be evaluated properly by looking at the introductory remarks to his main writings, e.g. *The Science of Mechanics, The Principles of the Theory of Heat*, etc. In these texts, the famous anti-metaphysical aim of Mach's epistemological concerns is expressed in a way that reveals its place within a well-defined philosophical framework. In the preface to the first edition of *The Science of Mechanics*, for example, Mach declares that the aim of the volume is "to clear up ideas, expose the real significance of the matter, and get rid of metaphysical obscurities" (Mach 1919, ix). This "enlightening or antimetaphysical intention" (in the original German text, Mach speaks of "*eine aufklärende oder antimetaphysische Tendenz*") clarifies the goal of the entire book, which is to provide a historico-critical investigation into the principles of the science of mechanics. From the viewpoint of the history of philosophy, this is of some interest, for the outlined conceptual cluster represents an ideal development of the philosophical methodology that was standard in Mach's time. The name that he gives to his own approach clearly evokes Kantian *criticism*, one of the most important products of philosophical *enlightenment* (*Aufklärung*), but Mach's version claims to be original, especially since it aims to obliterate the metaphysical obscurities that Kant had been unable to abandon. Mach's criticism is in fact a *historical* enquiry that attempts to cast light on the nature of ordinary scientific concepts by tracing their origins through the obscure paths of past cultures and civilizations.[3]

The explicit contraposition with Kant is expressed especially clearly in the *Analysis of Sensations*. In this text, Mach famously admits the early influence that Kant's *Prolegomena to any Future Metaphysics* exerted on him, also remarking, however, that he soon rejected Kant's "idealistic standpoint" as a hypothesis that cannot play a consistent role in scientific inquiry.[4] In the preface to the fourth edition of that same book, Mach in fact upholds "the opinion […] that science ought to be confined to the compendious representation of the actual", claiming that this "necessarily involves as a consequence the elimination of all superfluous assumption which cannot be controlled by experience, and, above all, of all assumptions that are metaphysical in Kant's sense" (Mach 1959, xl). Therefore, it seems that getting rid of metaphysical content is not only a matter of empiricism, for Mach–something that Kant admitted as well, in his tentative redefinition of the boundaries of reason based on a recognition of the sensible character of human intuitions and the role it plays in ordinary knowledge (a redefinition that in fact built on earlier attempts to determine a new philosophical method by thinkers such as Descartes and Leibniz). Mach clearly tried to go one step further; he tried to proceed on the same path as classic philosophers, but also to leave them behind, in the end. For him, the Kantian idealistic and transcendental philoso-

[3] In his communication to the 2016 *Centenary Conference* on Ernst Mach held by the Institut Wiener Kreis, Don Howard argued that the reference to a historico-critical approach in the title of Mach's works may be a response to the nineteenth-century tradition of Biblical criticism and the reception of its humanistic and hermeneutic methods (reference to this can be found, e.g., in Patton 2019, 337; and Uebel 2019, 505). This is of course an intriguing viewpoint which deserves to be thoroughly explored. In her study of Mach's historico-critical method, for example, Elisabeth Nemeth (2019) critically discusses this idea, showing that it is possible to find a broader frame of reference for his approach to scientific inquiry, including the historicist attitude of the natural sciences of his time.

[4] Cf. Mach 1959, 30 fn., 361, 367, 368; Mach, 1910, 234. On Mach's relationship with Kant, see e.g. Blackmore 1972, 10–11, 289; and Banks 2003, 24–25.

phy, which is interested in "metaphysical cognitions" whose "sources cannot be empirical" (Kant 2004, 15), is as inadequate "as a guide for modern scientific research" (Mach 1959, 368) as any other rationalistic attempt that tries to grasp something that is believed to lie beyond empirical experience. But the further step Mach had in mind can be performed by applying the same methodological principle implied in the Kantian conception; that is to say, it may be possible to enlighten obscure metaphysical content by performing a *critical* analysis. On this, one can appreciate the originality of Mach's approach. Inspired by the "historical trend" characteristic of his cultural framework (cf. Haller 1988, 69; and Nemeth 2019, 28), Mach adopted historical studies as the only tool that might allow us to assess the value of scientific knowledge claims properly. For him, history in fact sheds light on the relative (i.e. situated) character of scientific concepts, theories, and laws, thus allowing for a reconsideration of the value we should attribute to them as a "truthful explanation" of facts. Making use of another Machian concept, we might say that a historico-critical approach shows us the purely *economic* value of scientific knowledge–which is in fact the fundamental epistemological principle of Mach's view.

Within this framework, the philosophical consequences of Mach's methodological tenets are especially interesting, for they allow us to outline a conception of the role and value of scientific knowledge which might be put into dialogue with contemporary approaches in the philosophy of science. In particular, these tenets may be seen as the foundation of a pragmatist epistemology focused on the situated character of scientific knowledge (which Mach famously conceives of as a development of human knowledge in general) and its dependence on a factual basis. Moreover, Mach's historico-critical method stresses the fallibilist nature of science, thus inviting us to give primary attention to the operational efficiency of our theories and ideas, i.e., to how well they help us to cope with our wealth of experience. In the following sub-sections, I will try to show the extent to which this pragmatist conception is supported by or can be built on Mach's epistemological concerns, with the secondary aim of providing further elements, in addition to the existing scholarship, with which to assess the relevance of Mach's approach to scientific knowledge.

The three important issues that characterize Mach's philosophical methodology ("enlightenment"; "historical studies"; "economy of (scientific) thought") have been variously explored in the scholarly literature on Mach. It is neither necessary nor of great interest to repeat what has already been thoroughly studied. Therefore, starting with a couple of recently published papers, I will touch on these issues only briefly, elaborating on them only as much as is necessary for the purposes of my argument.

1.1 Enlightenment

In the preface to the *Principles of the Theory of Heat*, Mach declares:

> The present work sets itself a similar task to that confronted in my *Mechanics.* It aims to give *a critical epistemological elucidation* of the foundations of the theory of heat, to lay out for inspection the facts that influenced the formation of the relevant concepts, and to show why and to what extent the former are to be understood in the light of the latter. […] By means of […] many of the individual discussions here collected […] I have some hopes of fulfilling my general aim: to eliminate idle and superfluous

> notions and unwarranted metaphysical assumptions from this branch of physics also. (Mach 1986, 1; emphasis added)

As he had in 1883, Mach defines his own assessment of scientific knowledge as an "*erkenntniskritische Aufklärung*", thus remarking the reference to the Kantian philosophical framework. He also reaffirms the general aim of this critical elucidation, which is supposed to allow us to obliterate any obscure, i.e. nonsensical and meaningless, conception from our own field of study. According to Philipp Frank, who as early as 1917 stressed the role played by Mach as a philosopher of enlightenment, it is precisely the "struggle against the misuse of auxiliary concepts"–that is, an attempt to show that concepts have a realm of validity outside of which they cannot be used properly–that characterizes the "Age of Enlightenment". In addition, Frank stresses the historically contingent value of the auxiliary concepts and argues that every period in the history of science tends to misuse the concepts inherited from the previous one; therefore, "in every period a new enlightenment is required in order to abolish this misuse. […] To this work Mach dedicated himself" (Frank 1970, 229, 231).

This issue has recently been explored by Thomas Uebel in an interesting study on Mach's "enlightenment pragmatism". First, Uebel appreciates the continuity between Mach's "critical empiricism" and "the Enlightenment ideals of which, in German-speaking lands, Kant was the best-known advocate" (Uebel 2021, 85). Second, he argues that for Mach, "enlightenment was […] an ongoing project involving many distinct episodes, edging uncertainly forwards" (Uebel 2021, 89–90). The most important consequence of this for Mach's view of scientific knowledge is that it implies that our world-description, and especially the scientific world-explanation, must be considered a continuous development whose individual moments, contextualized as they are within historical and/or cultural viewpoints, do not provide us with a complete or literally accurate reproduction of the concerned states of affairs. As much as any historical event, the scientific conceptualizations are in a state of becoming, and no correspondence of these abstract models with the facts they are supposed to reproduce can be assessed properly. In fact, Uebel also remarks that a distinctive point of Mach's enlightenment perspective resides in his advice that we "consult the contingent history of our very own projects and find their measure of success and failure in what we ourselves can discern" (Uebel 2021, 89). This is also significant, for it allows us to reflect on the extent to which and how Mach problematizes the traditional approach to scientific claims, which maintains that it may be possible for us to judge their value based on their adequacy in capturing states of affairs.

Indeed, the ideal of enlightenment is expressed in Mach's attempt to reassess the value of scientific knowledge based on anti-foundationalist principles.[5] To act critically on that issue is to face its nature and to attempt to grasp its proper meaning–which apparently is not grounded in a metaphysical essence that we can know–with no fear of dealing with a multifaceted and changeable subject. As Uebel remarks, for Mach the aim of a critical elucidation is precisely to bring scientific "conceptualisations, whether mere concepts or whole theories, back down to earth and make them available for critical discussion" (Uebel 2021, 88).[6]

[5] Paul Feyerabend (1970, 178) remarked that "Mach develops the outlines of a knowledge without foundations", and Rudolf Haller (1988) appreciated his naturalistic anti-foundationalism as well. On this, cf. also Uebel 2021, 92 ff.

[6] Since Mach conceived of metaphysics as "what we place beyond the reach of experience–indeed, far above it, what we cannot remember ever having learnt", including "ideas that have become absolutized" (Uebel

The final consideration of Uebel's paper that I find especially interesting concerns the relationship between the enlightenment and the pragmatist attitudes. Indeed, Uebel further argues that "the '*Selbstbesinnung*' [self-reflection] that enlightenment demanded required the naturalisation and pragmatisation [of reason]: […] to make a difference, reason had to adapt to the human scale" (Uebel 2021, 89). Therefore, it can be maintained that it is precisely the way Mach reaffirms the ideals of enlightenment within a new framework that allows us to assess the pragmatist or *humanistic* features of his view.[7] Furthermore, our appreciation of Mach's contribution to the philosophy of science should stress the role that the idea that scientific knowledge is situated plays in his critical epistemology. "What merits thinking of Mach as a pragmatist", Uebel continues, "is his insistence, as a philosopher, on the ultimately practical orientation of all thought as a matter both of fact and norm, and, as a historian of science, on the need to investigate the specific problem situations out of and in response to which concepts and theories developed" (Uebel 2021, 84). This is especially relevant to my argument since among the pragmatist (or proto-pragmatist) features we encounter in Mach, the situated character of human knowledge, and therefore of science, is shared by perspectivists such as Hasok Chang, for example (cf. Sect. 2).

From what has been seen thus far, it might be argued that for Mach, philosophical enlightenment allows us to focus on the contextual and supposedly non-foundational value of scientific knowledge. Mach's criticism defends the view that our concepts are not fixed at all; rather, they depend on the framework of reference out of which they arise and within which they are used, and their meaning may be modified, re-valued, or even rejected when that framework changes. But this view requires a new method, a new tool. This activity cannot be a transcendental enterprise aimed at encountering abstract principles of knowledge. On the contrary, it should be a procedure that allows us to appreciate the changing nature of our conceptualization and the importance of a pragmatic assessment of it based on how that conceptualization copes with the problems one aims to solve, from the specific perspective one takes on them. This procedure should be a *historical* one, in fact, since history seems to be the only device that enables us to approach, in a proper way, that "real living manifestation of the human race" that is science (von Mises 1987, 171; cf. also Uebel 2019, 504).[8]

1.2 Historical studies

The role played by history in Mach's critical elucidation is clearly expressed in the *Introduction* to the *Principles of the Theory of Heat*, where he offers a relativized and instrumen-

2021, 88), the critical elucidation he performs may be the only proper anti-metaphysical means philosophers can count on.

[7] These *humanistic* features must be read in light of Ferdinand C. Schiller's proto-pragmatist focus on the human viewpoint as the actual framework of our world-description. I hope to make this clearer in what follows.

[8] In his seminal study, Richard Von Mises further observes that "Mach recognized that the most reliable means of clarifying methodological questions was to study their history–to follow the growth of ideas as they developed in the minds of the great scientists" (von Mises 1987, 170). In my opinion, this observation does not capture the complexity and richness of Mach's approach to knowledge, which involves much more than the mere plane of the history of science as a discipline. By this I mean that Mach focuses on the development of ideas not only in the minds of the great scientists but in the overall history of mankind, with a special focus on how we (human beings) experience world events. Mach's "historical method" is therefore a proper tool for epistemological reflection insofar as it engages with both the biological and the civilizational foundations of our conceptual system (on this, cf. e.g. Nemeth 2019).

tal interpretation of the value of scientific knowledge, based on a contextualization of its conceptions:

> Historical studies are a very essential part of a scientific education. They acquaint us with other problems, other hypotheses, and other modes of viewing things, as well as with the facts and conditions of their origin, growth, and eventual decay. Under the pressure of other facts which formerly stood in the foreground other notions than those obtaining to-day were formed, other problems arose and found their solution, only to make way in their turn for the new ones that were to come after them. Once we have accustomed ourselves to regard our conceptions as merely a means for the attainment of definite ends, we shall not find it difficult to perform, in the given case, the necessary transformations in our own thought. (Mach 1986, 5)

These observations are consistent with the revaluation required by the enlightenment approach, which attempts a thorough analysis of the various elements that constitute our world-description. According to Mach's renewed critical method, this analysis must not be a logical enterprise, but rather a process of reconstructing the historical development of those elements. Such a process allows us to assess a discipline, for it "lets the current development of a certain science appear in an essentially contingent light, and thus prevents dogmatic solidification" (Michael Heidelberger and Wolfgang Reiter in Mach 2016, xiii). That is to say: history enlightens the conceptualizations that are ordinarily adopted and explores their origin in the biological, cultural, and technological history of mankind; contextualizing these conceptualizations, it reveals the situated character of scientific knowledge, and thus its relative value. The question is: how are we supposed to evaluate this relativization? Does Mach's assessment lead to a radical form of anti-realism, to a sceptical and sterile relativism that treats scientific concepts and theories as mere instruments for an abstract description of the experienced world, thus leaving us with no objective facts and no claims to universal validity at all? Or is a moderate realism of some sort–that is, a view which might combine realist features with instrumentalist or constructivist conceptions of scientific knowledge–maintained in Mach's critical empiricism? In what follows, I would like to offer a tentative answer to this.

It is well known that Mach's historical approach to epistemology traces back to his early writings, especially to the lecture on the *History and Root of the Principle of the Conservation of Energy* held in Prague in 1871 and published the following year. As we read in the preface to the second edition of that lecture (1909), Mach conceived of it as a "first attempt to give an adequate exposition of [his] epistemological standpoint [*erkenntniskritischen Standpunktes*] […] with respect to science as a whole" (Mach 1911, 9). In a recent paper, Luca Guzzardi outlined the way in which Mach expresses his epistemological point of view, remarking that for Mach, the lack of a sense of history when approaching "propositions which have often cost several thousand years' labour of thought" (Mach 1911, 17) may promote a reifying, metaphysical attitude, as far as "we are accustomed to call concepts metaphysical, if we have forgotten how we reached them" (Mach 1911, 17; cf. Guzzardi 2021, 167). "Let us not let go the guiding hand of history"–continues Mach (1911, 18)–for "history has made all; history can alter all". And "if from history one learned nothing else than the variability of views, it would be invaluable" (Mach 1911, 17). But why is this so? Why is history such a powerful and necessary critical tool? Because it stresses the situatedness of our conceptualization, both *diachronically* (different viewpoints or interpretations at different times) and *synchronically* (different viewpoints or interpretations at the same time). It shows that scientific ideas, concepts, theories, and laws are not literal expressions of the states of affairs but rather, first and foremost, the momentary products of an empirical activity of technical experiencing.[9]

As Erwin Hiebert stresses in his classic paper *Mach's Use of the History of Science*, "the conceptual products of science, always incomplete, take on a form at any particular time which reflects the historical circumstances and the focus of attention of the particular investigator–now physicist, now physiologist, now psychologist" (Hiebert 1970, 188-189). Hiebert further argues that, "with the passing of time, what is accidentally (historically) acquired is converted into the philosophically argued", and for that reason "it becomes mandatory for the scientist to recognize and counteract the insidious gradual process by which scientific constructs come to be regarded as philosophically necessary rather than historically contingent" (Hiebert 1970, 189). The danger that Mach dealt with resides in our "temptation to reify mental constructs like substance, force and energy. The study of the history of science reveals that such constructs, originally provisional, in time and by successful application receive metaphysical sanctions that are virtually unassailable" (Hiebert 1970, 202).

The problem at issue in Mach's works thus involves the actual value of scientific knowledge–that is, its "value of use" as a means for a fruitful world-representation, which is not necessarily an explanation of the relevant states of affairs in the traditional (metaphysical) sense. His critical reflections on the historically contingent character of scientific claims, their being situated, do not imply that we cannot use science consistently. That is, Mach does not reject the operational efficiency of scientific theoretical systems but only the ordinary belief that they can provide adequate descriptions of events occurring in the world. For that reason, such a clarification is of the greatest importance for Mach, for it allows us to rid science of metaphysical obscurities that may be detrimental to the development of future inquiries. Thus, the history of science can have an important function insofar as it shows that ordinary scientific concepts result from a process of intellectual solidification that isolated them from their historical origin (cf. Guzzardi 2021, 171), and this implies a misinterpretation of their value, which Mach claims to be properly assessable only on a functional or operational basis.

This leads us to what can be seen as the core epistemological tenet of Mach's view, namely the economical nature of (scientific) thought. In writings such as *Popular Scientific Lectures* and *Knowledge and Error*, as well as important chapters from the *Principles of the Theory of Heat*, Mach repeatedly maintains that conceptualization is in fact a collection of intellectual instruments that "allow us to find our way in the bewildering tangle of facts" (Mach 1976, 98) and whose actual role is to "provide the fully developed human individual with as perfect a means of orientating himself as possible" (Mach 1959, 37). For him, our conceptual system does not reproduce the explored events adequately, for it is only an abstract description of them. But it is worth reflecting further on Mach's endorsement of this anti-realist view, especially if one considers that the relationship between facts and mental symbols that he defends is not as obvious as it may first appear. As I will try to show,

9 In this paper, I focus on the theoretical side of Mach's epistemology, which was of course based on his experimental works and scientific practice. On this relationship, see e.g. the relevant studies provided by Alexandra Hui (2013; 2021) and Richard Staley (2017; 2018).

a careful inquiry into the nature of this relationship in Mach's thought can provide us with relevant information for the argument I would like to defend in the second part of this paper.

1.3 Moderate constructivism

Alongside Mach's anti-metaphysical aim, the issue of economy of thought is also well known; it is therefore unnecessary to explore it thoroughly.[10] On the contrary, the relationship between facts and theories in Mach has received less attention, and for this reason it is still a worthwhile topic of study.[11] I made the case for a moderate constructivist reading of this relationship in a previous paper (Gori 2021). As I tried to argue, it is possible to say that Mach conceived of scientific knowledge as an ideal attempt to mirror facts in thought that presupposes a dynamic yet empirically grounded relationship between theories and facts. Given that theories must always respect facts, Mach reflected on how theories *reconstruct* facts in thought by claiming that this process may determine an extension in the range of the facts explored, thus revealing new, previously unobserved features or even new facts to be further explained and accommodated within the theory (cf. e.g. Mach 1895, 233; 1976, 98).[12] Thus, the relationship between theories and facts is dynamic not because it is "a dialectical process that transforms both ingredients", as argued by Paul Feyerabend (1984; 8–9; cf. on this Preston, forth.), but because it implies that an assessment of the mental-economical activity of reconstructing facts in thought cannot be performed in light of the ordinary conception of truth, which defends the idea of a literal representation of facts through scientific theories. Rather, Mach adopts the evolutionary model of an *adaptation of ideas to facts*, which is especially helpful for interpreting his view properly, for it allows us to maintain both a moderate realism of a pragmatist kind (the idea that theories constantly expect confirmation from facts and that this might lead to the *refinement* of the scientific world-representation, allowing for a "purer expression of the facts"; cf. e.g. Mach 1910, 230; 1911, 57; 1919, 514; Uebel 2021, 97 ff.) and a constructivist conception that is compatible with a perspectival interpretation of the entire process of scientific explanation (the idea that what is in fact relevant to scientists is the class of facts that the theoretical activity allows them to deal with–that is, the idea that how *theories* are constructed is crucial, for this determines their access to facts; cf. Mach 1976, 102).

Let me try to elaborate what I would like to defend. What I aimed to argue in my 2021 paper is that in Mach we find the seeds of a *modest realist* view which both maintains that the goal of science is to find out how world events really occur and asserts that scientists are making progress in this direction but which also recognizes that this goal will never be fully achieved insofar as science can only provide us with an indirect description of states

[10] Lydia Patton (2021) has recently dealt with Mach's conception of the economy of science, providing interesting remarks on how "abstraction, pragmatism, and history work together" in his epistemological view.

[11] The issue of what a fact is, for Mach, has only been addressed indirectly, e.g. by John Blackmore (1972, 32) and Gerald Holton (1988, 247), but a thorough study on the subject matter was recently published by De Waal and Ten Hagen. In their paper, De Waal and Ten Hagen observe that "Mach's reputation as a positivist only interested in facts is an oversimplification" of his view. They maintain that "Mach made a clear distinction between fact and theory, but he also acknowledged that the two crucially depended on one another" (De Waal and Ten Hagen 2020, 65).

[12] This, of course, would determine the necessity of developing an updated version of the old theory, or even replacing it completely with a new one.

of affairs.[13] This view combines a realist tenet concerning the *goal* of scientific inquiry with an instrumentalist conception of scientific claims, presupposing an essential and constant interaction between the knowing subject and the world events that we experience. It therefore seems to be consistent with the sort of realism implied in perspectivist views such as that offered by Ronald Giere, who, for example, argues that

> the stronger claims a scientist can legitimately make are of a qualified, conditional form: "According to this highly confirmed theory (or reliable instrument), the world seems to be roughly such and such". There is no way legitimately to take the further objectivist step and declare unconditionally: "This theory (or instrument) provides us with a complete and literally correct picture of the world itself" (Giere 2006, 5–6).[14]

In my view, it is possible to find in Mach a position that is consistent with *this* sort of realism, that is, a view that rejects any defence of objectivist tenets while reflecting on the relationship between theoretical and observation statements in a way that calls for a problematization of that relationship, as if Mach were not completely satisfied with strong instrumentalist or constructivist stances that leave aside the ideal of mirroring facts in thoughts, which he apparently never abandoned (cf. e.g. Mach 1895, 186; 1986, 361). Accordingly, in Gori (2021) I argued that it may be possible to ascribe to Mach a *moderate constructivist* tenet which does not pretend to grasp the ontological level of our world-representation but that is limited to the inaccurate, albeit operationally efficient, mental reconstruction of facts of which we are capable (e.g. Mach 1976, 355).[15]

This can be appreciated if we look at how Mach deals with the transformation of scientific thoughts and ideas, and their adaptation to facts. In a series of papers and book chapters (e.g. Mach 1986, Ch. 25; 1976, Ch. 10), Mach argues that conceptualization is a product of the mental-economical (*denkökonomisch*) activity of conceptual completion of facts in thought, which, according to Mach, especially characterizes the scientific world-description. This

[13] In *Principles of the Theory of Heat*, Mach observes that "what we call a 'theory' or a 'theoretical idea' which is the starting point of a theory, falls into the category of indirect description [*indirekte Beschreibung*]", that is, "a description in which we make use to a certain extent of one already given elsewhere or even of one that has yet to be worked out accurately" (Mach 1986, 365).

[14] I will say more on this in Sect. 1.4. In his seminal work on perspectival realism, Ronald Giere provides us with a definition of "moderate constructivism" which corresponds to the view of modest realism I have outlined above. In fact, Giere's *perspectival* realism is a development of what he earlier called *constructivist* realism (cf. Giere 2006, 11, 118). Another interesting reference for reflecting on a more "flexible" approach to scientific knowledge which is supposed to "steer a course between scientific realism and instrumentalism" is Mary Hesse, whose "consensus theory" focuses on the interplay between the theoretical and the observational activity and problematizes the idea that the world structure which scientists ordinarily claim to exist "can ever be accurately *known* or represented in language" (Hesse 1980, xiv, 446; on this cf. Gori 2021, 392 ff.).

[15] On this I tend to diverge from both strong anti-realist interpretations of Mach focused exclusively on his interest in the constructed task of human knowledge (cf. e.g. Uebel 2019, 506) and readings claiming that Mach attempted to provide us with a robust realist world-description (cf. e.g. Banks 2004; 2014). Although I find Eric Banks's interpretation fascinating and grounded in good textual evidence, I am inclined to maintain a cautious reading of Mach's defence of direct realism about world-elements, given the number of passages in which Mach refuses to endorse any sort of metaphysical commitment (whether realist or idealist) and reiterates his interest in the plane of the *experienced* facts as the only realm of which one can speak properly. I have also argued that it may be possible to interpret Mach's view as an "epistemological agnosticism", for I believe that, in the end, Mach was not interested in investigations that try to grasp what cannot be expressed by meaningful knowledge claims (cf. Gori 2018; 2021).

view is also explored in an important paper published in 1892, *Facts and Mental Symbols*. For Mach, in dealing with facts, scientists "mentally *supply*" them with economical symbols (Mach 1892, 200), i.e. they put "in the place of a fact something *different*, something more simple, which is qualified to represent it in some *certain* aspect, but for the very reason that it is different does not represent it in other aspects" (Mach 1892, 201). Therefore, it can be argued that "our theories are abstractions, which place in relief what is important for certain fixed cases, but also neglect almost necessarily, or even disguise, what is important for other cases" (Mach 1892, 201). This does not mean that theories create facts, of course. At the same time, however, Mach seems to make room for the idea that it is impossible for us to directly mirror observed phenomena in thought and that we instead ordinarily work with "models" that only "imitate in form" actually occurring events (Mach 1892, 202). Anticipating an observation that he would go on to elaborate further in other works, Mach argues that "we have to distinguish sharply between our theoretical conceptions of phenomena and that which we observe. The former must be regarded merely as auxiliary instruments that have been created for a definite purpose and which possess permanent value only with respect to that purpose" (1892, 202). Thus, the question at stake seems to be this: as scientists, we live in a world of theoretical elaborations that reproduce natural events in the form of mental-economical symbols; our theoretical statements are rooted in observational statements, however, and they somehow and to some extent correspond to what Mach calls "sensible reality" (Mach 1976, 356). How is this so? And what is the actual value of the scientific world-representation, for Mach?

Throughout his writings, Mach repeatedly stresses that science is a process of *adapting thoughts to facts and thoughts to each other* (cf. Mach 1895, Ch. 12; 1986, Ch. 25; Mach 1976, Ch. 10) and devotes highly interesting sections of his works to explaining how this adaptation must be interpreted. As is well known, Mach maintains that "knowledge is a product of organic nature, and […] if Darwin reasoned rightly, the general imprint of evolution and transformation must be noticeable in ideas also" (Mach 1895, 217-218). Given that science is simply a highly elaborated version of human knowledge on Mach's view, this can also be applied to theories and laws of nature. The way in which Mach deals with this issue allows us to argue that he adhered to a strong form of evolutionary epistemology which maintains that it is possible to "account for the characteristics of cognitive mechanisms in animals and humans by a straightforward extension of the biological theory of evolution to those aspects or traits of animals which are the biological substrates of cognitive activity", and that the same can be done "for the evolution of ideas, scientific theories and culture in general by using models and metaphors drawn from evolutionary biology" (Bradie 1986, 401-403).[16]

If this is the case, it is possible to infer something interesting about the relationship between thoughts and facts in Mach's works. On the one hand, there is a system of thoughts, ideas, theories, etc., that must make up a coherent whole. Their survival within this system is determined by their ability to agree with one another and to contribute to mutual support. On the other hand, there is the environment to which thoughts must adapt, i.e. the domain of facts to be explained or described. If the analogy with natural selection must be understood in a non-metaphorical sense, it might be said that a mutual adaptation is involved, for the

[16] On Mach's commitment to an evolutionary epistemology–or, less radically, to a biological conception of knowledge–cf. Čapek 1968; Haller 1988; and Pojman 2011. For a contextualization of this conception in Mach's intellectual framework, see also von Mises 1987; Wolters in Mach 1985, xiii; and Goeres 2004, 46 ff.

Darwinian model implies that the environment constantly changes in compliance with the biological modifications of the species it hosts. This would suggest that Mach held a strong constructivist view, according to which thoughts determine a modification of facts themselves. However, no evidence for this can be found in Mach. On the contrary, he repeatedly maintains that ideas are rooted in facts and that it is thoughts that adapt to facts, not the reverse. At the same time, Mach conceives of theoretical activity as determining an extension of the domain of facts to be explained–not of facts themselves, but of the "constantly widening sphere of action" of ideas, the transformation of which "appears as a part of the general evolution of life" (Mach 1895, 233).

Just how this type of modification of the environment should be understood can be clarified on the basis of Mach's biological conception of knowledge. As Mach argues in *Knowledge and Error*, "scientific thought arises out of popular thought, and so completes the continuous series of biological development that begins with the first simple manifestation of life" (Mach 1976, 1). For him, scientific knowledge is but a highly developed attempt to mentally supply incomplete observational findings, and the concepts it elaborates "consist in consciousness tied to a word of the reactions to be expected from the class of objects or facts denoted. […] The object corresponds to the concept, if it yields the expected reaction when tested in the way intended" (Mach 1976, 97; cf. also 93). Although concepts are "rooted in facts" (Mach 1976, 99) and always have a "factual correlative" (Mach 1976, 102), Mach argues that the agreement between our mental symbols and the results of observation should be evaluated *pragmatically*, as a matter of the response elicited by our attempts at explanation.[17] In fact, Mach defines scientific knowledge as "a mental experience directly or indirectly beneficial to us" that "flows from the same mental source" as error, and further argues that "only success can tell the one from the other" (Mach 1976, 84). What is relevant to evaluating our judgements, then, is only whether they "stand up" (1976, 84) and allow us to achieve the expected result.

This revaluation of the idea of "correspondence" on a falsificational basis, which leads to the view that we can only assess the truth value of theories and ideas *negatively*, i.e. by showing that they are "not false", should be combined with Mach's conception of scientific progress. Indeed, in the *Popular Scientific Lectures* Mach observes that "experience never ceases, and science, accordingly, stands midway in the evolutionary process" of adaptation that determines ever "richer" (i.e. more complete and therefore adequate) conceptions of facts (Mach 1895, 227). Similarly, in *Knowledge and Error* he remarks that "more accurate quantitative enquiry aims at determining facts as completely as possible", and that "the progressive refinement of the laws of nature and the increasing restriction of expectations corresponds to a more precise adaptation of thought to fact" (Mach 1976, 355). Mach's "falsificationism" can therefore be defined accordingly: by "eliminating" the theories and ideas that are inconsistent with the observed facts, we are left with the "fittest" ones, and this process allows us to *refine scientific knowledge* towards a "richer" and "purer" world-representation (cf. Mach 1910, 230). But the latter will always remain a representation, of course; i.e., it will always be nothing more than an indirect description of states of affairs (cf. Mach 1986, 365). Mach indeed admits that "it is not possible to achieve perfect adaptation to every individual and incalculable future fact" (Mach 1976, 355), and in the 1894

[17] The pragmatist features of Mach's epistemology are explored, e.g., in Uebel 2019; 2021. A broader account of pragmatism within the philosophical framework of logical empiricism is provided in Pihlström, Weidtmann and Stadler 2017.

paper *On the Principle of the Conservation of Energy* he also argues that "it remains an open question how far nature corresponds to […] our formal need of a very simple, palpable, substantial conception of the processes in our environment […] or how far we can satisfy it" (Mach 1894, 54).

1.4 Pragmatic instrumentalism

At this point, an apparent tension between two views of scientific knowledge seems to arise–one which exemplifies the modest realism or moderate constructivism outlined by Ronald Giere. On the one hand, we find in Mach the idea that the human mind "attempts to mirror in itself the rich life of the world" (Mach 1895, 187; cf. also Mach 1986, 361), which is consistent with the ideal of mental adaptation to facts as an attempt to reproduce observed phenomena as adequately as possible. On the other hand, Mach conceives of science as a mere tool for orientating ourselves in the natural world (Mach 1959, 37; Mach 1976, 2, 98, 354). For him, the elaboration of conceptual symbols allows us to spare mental effort and to reproduce a large number of facts in thought (Mach 1895, 191 ff.), but this activity implies an intervention into the received data and, consequently, results in a contraposition between theoretical and observational statements (cf. Mach 1986, 365).

What remains uncontroversial (I hope) is that Mach does not defend an instrumentalism that leads to a sterile form of sceptical relativism–and this is the case precisely because of his pragmatist attitude, which embraces fallibilism and calls for a redetermination of the basic tenets of and key notions involved in scientific explanation ("knowledge", "experience", "truth", etc.; on this, cf. e.g. Putnam 1995, 21; Uebel 2021, 99). Mach indeed assumes that the "potential knowledge" that concepts contain (Mach 1976, 83) can be properly assessed on the basis of how well they help us to "cope with the wealth of experience" (Mach 1976, 81). In principle, "the course of representations should adapt itself as closely as possible to physical and psychical experience"; it "is to be as faithful a picture as possible of the course of nature herself" (Mach 1976, 81–82). This last remark is especially relevant, for Mach refers precisely to states of affairs and leaves aside the utilitarian language focused on the operational efficiency of theories, talking instead of a "faithful" (*getreues*) picture of the course of nature (*Naturlauf*). I agree that Mach rejects objective realism as a viable option, given the fundamentally economical character of our theoretical relationship with world-events, and yet he seems to maintain the idea of the progressive development of science towards a literal description of those events–at least (or at most) as a regulative ideal. As I will try to show in the next section, this could be enough to allow us to reflect on the consistency of his view with Giere's, who also endorses a fallibilist interpretation of scientific claims and argues that "perspectival realism is *as much realism as science can provide*" (Giere 2006, 16, emphasis added; see Sect. 2).

In addition to this, it may also be worth considering that Mach never abandons the notion that our engagement with the world is grounded in the physiological plane. Accordingly, he argues that "while concepts do indeed not exist as physical 'things', our reactions to objects [i.e. facts] of the same class of concept are psycho-physically similar,[18] and those to objects

[18] According to Eric Banks (2004, 43), "Mach expected the result of advances in sense physiology to be that the world elements of nature might become as directly accessible to human beings as their own sensations of colour or sound". To appreciate Mach's conception of direct knowledge, it may be worth looking at the experimental work which provides the background of his epistemological writings. Interesting work on this

of different classes dissimilar" (Mach 1976, 99). Of course, this does not mean that the concepts elaborated by science do correspond to something independent of the inquirer. At the same time, however, we might read this remark as an attempt to moderate an anti-realist constructivist view by focusing on a more fundamental (and perhaps common) plane than that of theories themselves. Although for Mach it is always a question of how efficiently science can elaborate our stimuli into a coherent system, it might be argued that those stimuli themselves are grounded in a further plane which lies at the basis of our experience as human beings. Of that world which "exists only one" (Mach 1895, 199) we can likely say nothing meaningful, and yet it still limits the proliferation of inconsistent representations, whose value is not merely a matter of coherence but depends on the pragmatic compliance of our mental symbols with the facts they reconstruct in thought (cf. Mach 1895, 227).[19]

Further interesting aspects of Mach's pragmatist constructivism can be found, e.g., in the *Popular Scientific Lectures*, where scientific explanation is defined as the attempt to "comprehend without effort" a province of fact in order to "no longer feel lost in this province" and Mach argues that this goal is achieved once we "mentally construct the whole province" (Mach 1895, 194). Furthermore, in the interesting (albeit largely neglected) chapter of the *Principles of the Theory of Heat* titled "Comparison as a Scientific Principle", Mach maintains that "description" in science "is a construction of facts in thought" (Mach 1986, 370) and that "the strict definition of a concept and, in case it is familiar, even the name of the concept is a stimulus to a precisely determined though often complicated, testing, comparing or constructing *activity* whose result, in most cases perceptible by the senses, is a term in the extension of the concept" (Mach 1986, 369). In agreement with the pragmatic conception of knowledge introduced above, Mach finally argues that "the concept is not a finished idea, but [a] body of directions for testing some actually existing idea with respect to certain properties, or of constructing some idea from given properties. The definition of the concept, or the name of the concept, releases a definite activity, a definite reaction, which has a definite result" (Mach 1986, 381). According to what has been said, Mach argues that a constructive activity takes place in thought, while admitting that concepts are rooted in facts. Therefore, we cannot mentally modify the latter on the ontological plane. What is remarkable, however, is that Mach thinks that scientific knowledge actually takes place at the mental-economical level, and for this reason it is crucial, on his view, to focus on this constructive activity. By this I mean that it is thought that provides us with access to facts and that we can therefore only talk about conceptualized objects or facts. Thus, it might be argued that what is actually relevant to scientists on Mach's view is the class of facts that the theoretical activity allows them to deal with.[20]

On this, it may be worth considering a couple of passages in which Mach focuses on the relationship between theoretical and observational statements. In *Principles of the Theory of*

has been done, e.g., by Paul Pojman (2011) and Richard Staley (2017; 2018). On Mach's idea that "psychology and psychophysics [were] well on their way to becoming fully exact doctrines", and on his view of precision and progress in science, cf. also Hui 2021.

[19] According to Sami Pihlström, this is the core of a pragmatist approach to scientific realism which attempts to moderate the tension between instrumentalism and a robust form of ontological realism. On this, cf. Pihlström 2007; 2009. For a discussion of the consistency of Mach's epistemology with pragmatic realism, cf. Gori 2018.

[20] On this, cf. Mach 1976, 102: "Thought does not occupy itself with things as they are in themselves, but with our concept of them; we know things only through their relations with other things." As I will argue in the final part of this paper, statements of this sort allow us to compare Mach's view with Hasok Chang's.

Heat, Mach argues that "a theoretical idea […] seem[s] to us to stand higher than the mere adherence to a fact or an observation [because it] can, and is intended to, present more than we see at the moment in the new fact; it can widen and enrich this fact with new features for which we are stimulated to seek–and which are often found" (Mach 1986, 365). Thus, a theoretical idea determines "an extension of knowledge which gives a quantitative advantage over mere observation" (1986, 365). Accordingly, in *Knowledge and Error*, Mach observes that concepts, whose purpose is "to allow us to find our way in the bewildering tangle of facts […] can represent and symbolize in thought large areas of fact. […] By bridging it under a concept, we simplify a fact, by leaving out of account those factual features that are irrelevant to our purpose, while at the same time amplifying the fact, by including all the characteristics of the class" (Mach 1976, 98). These excerpts outline a conception that is interested in the way in which our theoretical activity continuously results in a modification of how the phenomena to be explained are represented by us, both extending their range and focusing on different or new features. But this activity is only the first part of a process in which we elaborate hypotheses that must be successfully tested against sensible reality in order to become actual "knowledge", and it is precisely how they cope with their factual correlative that determines their meaningfulness for scientific inquiry.

The factual correlative of concepts can never be neglected, and according to Mach it is *not* subject to any modification by the concepts themselves.[21] Nevertheless, this process seems to imply some perspectival features, insofar as Mach conceives of scientific knowledge as situated within collective (e.g. historical, cultural, etc.) and/or individual frameworks. This is especially clear in those passages where Mach comments on the value of the mental symbols as *abstractions*. As shown above, in his 1892 paper he argues that "our theories are abstractions which place in relief what is important for certain fixed cases, but also neglect almost necessarily […] what is important for other cases" (Mach 1892, 201). Most importantly, it is only the *definite purpose* for which our "theoretical conceptions of phenomena" have been created that determines their value (1892, 201). Accordingly, in *The Science of Mechanics* we read that "in the reproduction of facts in thought, we never reproduce the facts in full, but only that side of them which is important to us, moved to this directly or indirectly by a practical interest" (Mach 1919, 482). Furthermore, in the *Popular Scientific Lectures*, Mach conceives of natural laws as "descriptions, that is, a mimetic reproduction of facts in thought", arguing that "the law always contains less than the fact itself, because it does not reproduce the fact as a whole, but only in that aspect which is important for us" (Mach 1895, 193). Finally, in *Knowledge and Error* Mach affirms that "facts are always somewhat arbitrarily and forcibly defined with a view to the momentary intellectual aim" (Mach 1976, 3), and he couches scientific knowledge in the historical development of human culture and civilization, as he had already suggested in the 1872 paper. But this perspectivization of the value of scientific knowledge does not pertain only to the historically oriented approach taken by Mach. In fact, it can also be encountered in the *analytical* criticism of the prejudices of scientists that he defends in *The Analysis of Sensations* and reaffirms elsewhere.[22] On more than one occasion, Mach indeed takes special care

[21] Of course, facts are not the elements; rather, they already pertain to a perspectival viewpoint. At the same time, however, facts are the "environment" our theories should adapt to and keep pace with. They are the "reality" our scientific claims actually talk about.

[22] On the consistency between the two anti-metaphysical approaches that can be encountered in Mach, see Guzzardi 2021, 171.

to remark that the dichotomy between the *physical* and the *psychical* is the expression of two different viewpoints describing the same event, and therefore that in calling the world-elements either "sensations" or "phenomena", we provide an inadequate interpretation of the state of affairs, for "in either terms an arbitrary, one-sided theory is embodied" (Mach 1895, 208–209).[23]

We can conclusively say that the historico-critical approach that is peculiar to Mach's methodology and his "philosophy of science" reveals the situated character of scientific knowledge and calls for a reconsideration of its value. The principle of the economy of thought combined with the idea of an ongoing adaptation of ideas to facts implies a pragmatist instrumentalism that allows Mach to avoid both scepticism and solipsism and to defend a sort of moderate realism which might be interpreted as a view that combines constructivist instrumentalism with the realist tenet that the *goal* of science is to explain facts, i.e., to describe them literally or directly. It at least seems arguable that Mach concerns himself with the tension between these approaches and that he tries to reflect on them from what can be defined as a pragmatist viewpoint, thus contributing significantly to the development of a modern approach to scientific knowledge. In the next section, I will argue that this view is to some extent consistent with perspectival realism, given the features that have been outlined thus far, and that this may be of some use in further assessing Mach's role in the history of the philosophy of science.

[23] It is important to deal carefully with the value Mach ascribed to sensations and their relationship with the elements. Although he repeatedly argues that the world we can know and meaningfully describe actually consists in our sensations (Mach 1959, 12), it is worth remarking that, for him, "sensations are [only] a special type of elements" (Gereon Wolters in Mach 2008, xvii) and that according to Mach's neutral stance, "it is misleading to conceive of elements as 'conscious contents' [*Bewusstsinsinhalte*]", for they are a far more fundamental entity. In fact, they are *Urphänomene* (archetypal phenomena), which can be perceived as sensations depending on the way we look at them (Goeres 2004, 50). Actually, given the three classes of elements that Mach famously describes–"world elements" A B C; "body elements" K L M; "intra-psychical" elements α β γ–only the functional relationship between the first and the second classes can be seen as a compound of "sensations" (Mach 1959, 16). When Mach agrees to "call *all* elements, insofar as we regard them as dependent on […] our body, *sensations*", he seems to explicitly endorse a perspectival view (Mach 1895, 209), for he maintains that when we speak of "sensations", we project a human viewpoint onto the elements, i.e. the *neutral* substrate of our relationship with the world. On this, see especially the first chapter of *The Analysis of Sensations* (e.g. Mach 1959, 16, 17 fn., 22).The way in which Mach considers the elements is also a problematic issue, for his texts allow for different interpretations. For example, one might argue, with Eric Banks (e.g. Banks 2014; 2021), that Mach defended a direct realist conception. Or, given that Mach claimed to be sympathetic with "the representatives of a philosophy of immanence" (especially Schuppe; Mach 1959, 46) and given that immanentist philosophers were commonly regarded as supporters of an idealist epistemology (cf. e.g. Sass 1976, 237), there may be space for an idealist interpretation of Mach's neutral monism. But Mach also declared his aversion to any attempt to accuse him of "idealism, Berkeleianism" and other '-isms' (Mach 1959, 48, 361), and he tried to avoid the realist as much as the idealist conception (cf. Mach 1959, 56). I am unable to deal with this issue here, for it would lead me far beyond the aim and scope of the present paper. Let me just say that a possible viable approach to it might start with a contextualization of Mach's view. That is, if we consider that Mach's criticism was oriented towards *materialistic* realism, on the one hand, and towards the *solipsistic* tendencies that follow from idealism, on the other (cf. Goeres 2004, 45, 54), there seems to be space for a more nuanced reading of the basic tenets of his epistemology.

2 Situated knowledge claims

One of the most interesting excerpts for considering Mach's perspectival view of scientific knowledge is a 1910 paper in which he "briefly represents the epistemology [*Erkenntnislehre*] to which [he has] devoted a good part of [his] life", with a special focus on his biological-economical view of an "*adaptation of thoughts to facts and of ideas to each other*" (Mach 1910, 225). In dealing with the critical remarks he received from Max Plank, Mach observes:

> Planck asks how the physical world-picture brought about by the application of these scientific methods came into existence. "Is the same merely a convenient but at bottom arbitrary creation of our mind, or do we find ourselves driven to the opposite conception that it mirrors natural processes which are entirely independent from us?" I can find *no irreconcilable alternative* here. *Convenient* it must be in order to guide us. How could we otherwise begin to do anything? But since it is also *dependent* on human individuality, it must also be at least in a certain sense arbitrary. […] Who can stop scientists from directing special attention to different sides of the facts? Perchance the decree of a sufficiently eminent physicist? But naturally the *human* world picture which *is socially preserved* by scientists succeeding or replacing each other will be more independent of individuality, progressively giving a purer expression of the facts. But in general, the observer comes into the picture in every observation and every opinion as does the environment as well. (Mach 1910, 230)

Quite significantly, Mach expounds the relevant consequences of what was explored in the previous section. In particular, he seems to combine an instrumentalist and (at least) mild constructivist conception of scientific knowledge with the modest realist tenet that our theories aim at a "purer expression of the facts". The scientific world-picture is a *human* and a *social* one; it depends on human individuality, and therefore every observation is perspectival at its very core. But at the same time, a positivistic ideal of scientific progress is maintained, this ideal affirming that the independence of the scientific world-description from the individual viewpoint may be possible, at a certain stage (although Mach remains quite sceptical about it). Let us consider the extent to which this view can be put into dialogue with the recent defence of perspectival realism.

In his seminal book *Scientific Perspectivism*, Ronald Giere makes the case for "an understanding of scientific claims that mediates between the strong objectivism of most scientists or the hard realism of many philosophers of science and the constructivism found largely among historians and sociologists of science", with the purpose of doing justice "to the constructivist critique of science […] while avoiding the excesses" (Giere 2006, 3). At the basis of the latter conception, Giere finds the view "that the process of doing science is so infused with all sorts of human judgements and values that what ends up being *proclaimed* to be the structure of reality may bear little resemblance to the real structure of the world" (Giere 2006, 7). For Giere, this does not in principle imply a denial that the world has a definite "objective" structure or that reality must be in some sense "constituted" by human activities.

Thus, a more nuanced conception might be defended, merging the relativism implied by the perspectival view with realist tenets of some sort.[24]

For Giere, "perspectivism makes room for constructivist influences in any scientific investigation" insofar as it maintains that "all theoretical claims remain perspectival in that they apply only to aspects of the world and then, in part because they apply only to some aspects of the world, never with complete precision. The result will be an account of science that brings observation and theory, perception and conception, closer together than they have seemed in objectivist accounts" (Giere 2006, 15). Furthermore, Giere stresses that "we simply cannot transcend our human perspective, however much some may aspire to a God's eye view of the universe" (2006, 15); it is indeed undeniable that science is a human activity and that "scientific instruments and theories are human creations" (2006, 15). Once we admit this twofold perspectivism concerning scientific knowledge, what needs to be done is to "show in detail *how* the actual practice of science limits the claims scientists can legitimately make about the universe" (2006, 15) and to consider whether there is still room for truth and objectivity within this picture. Giere answers this question by arguing that, provided that "some degree of contingency is always present in science", it might be maintained that "scientific claims are neither as objective as objectivist realists think nor as socially determined as even moderate constructivists often claim" (2006, 16). Thus, he argues that the constructivists seem to be "at least partly right", while "objectivist realism cannot be even an ideal goal"; consequently, "perspectival realism is as much realism as science can provide" (2006, 16). Indeed, it is important to remark that Giere's overall commitment is a *moderate form of constructivism* that affirms that "one could, in principle, be an objective realist about the *goal* of scientific claims, but then go on to argue that, as a matter of empirical fact, given the nature of human beings and human society, this ideal is not attainable" (2006, 11). From this he elaborates his original perspectival (modest) realist view, arguing that "the stronger claim a scientist can legitimately make is of a conditional form" and that, based on a "highly confirmed theory", one can only infer that "the world seems to be roughly such and such", and nothing more (2006, 5–6).[25]

It is not my intention either to provide a thorough reconstruction of Giere's interesting view or to discuss its details. Instead, I would like to stress the way in which he further elaborates the issue of his contingency thesis, i.e. the view that "scientific knowledge claims are perspectival rather than absolutely objective" (Giere 2006, 93), given the relevance this might have for the aims of the present paper. In order to defend his view, Giere observes the following:

> That human observation is perspectival, a function of an interaction between the world and human cognitive capacities, seems to me indisputable. And that the instruments that now dominate scientific observation are similarly perspectival seems almost equally indisputable. They are designed to interact selectively with the world

[24] In fact, Giere (2006, 13) makes it especially clear that "a *scientific* perspectivism does not degenerate into a silly relativism", and he tries to avoid the paradoxical, extreme remark that "every perspective is regarded as [being as] good as any other". His argument for a defence of perspectivism as a tenable thesis can be found, e.g., in Giere 2006, 81–82.

[25] Incidentally, the core of Giere's defence of perspectival realism relies on a reflection on colour vision. For Giere (2006, 14), "colours are *real enough*" but at the same time perspectival, and he especially stresses that they arise from the interaction between physical properties and our sense physiology (cf. e.g. Giere 2006, 31–32).

> in ways determined by human purposes. That the development of theoretical principles and abstract models is in part a function of past knowledge and current interests and ideas is less obvious, but, on reflection, seems clearly correct. Still less obvious is the conclusion that all the previous contingencies are not eliminated by the application of sound experimental methods. Here we must first realize that experimentation can only involve the meshing of several perspectives, some observational and some theoretical. But a combination of perspectives remains perspectival, not objective in some stronger sense. And even within perspectives, the strongest possible conclusion is that some model provides a good but never perfect fit to aspects of the world. A supposed "absolute conception of the world" (Williams 1985) is unattainable. This leaves us with a viable notion of scientific realism, but it is perspectival and contingent, not objective in some more absolute (metaphysical) sense. (Giere 2006, 93)

Giere's observations already provide interesting elements that might allow us to tentatively reflect on the consistency between perspectival realism and Mach's view. The role played by observation and theoretical elaboration in determining a perspectival view of states of affairs and the situatedness–within a framework determined by past knowledge as much as by current interests and ideas–of the technical apparatus adopted to perform these observations as well as of the theoretical principles and of the abstract models, resemble considerations that might be encountered in Mach (of course, contextualized within his own epistemological perspective). The most significant aspect rests especially on how the very idea of "realism" may be interpreted within this context. Indeed, both Mach and Giere seem to reflect on a sort of realism that is not objective and is only concerned with the conditional access to world-events that *we human beings* can secure.[26] But there is more to this, I think, and further considerations are needed before we can attempt to draw conclusive remarks.

Building on Giere's work, Michela Massimi provides a remarkable defence of perspectival realism, attempting to outline a new conception of *perspectival truth*. She takes perspectivism to be "first and foremost an *epistemic view* about the nature of our scientific knowledge", affirming that this knowledge is *situated* (Massimi 2018a, 164). Being situated is understood in two main ways:

> (1) Our scientific knowledge is *historically situated*, that is, it is the inevitable product of the historical period to which those scientific representations, modeling practices, data gathering, and scientific theories belong.
> And/Or
> (2) Our scientific knowledge is *culturally situated*, that is, it is the inevitable product of the prevailing cultural tradition in which those scientific representations, modeling practices, data gathering, and scientific theories were formulated. (Massimi 2018a, 164)

[26] On this, it may be worth considering what Hans Kleinpeter–one of the early upholders of the realist interpretation of Mach's monism, quoted by Banks in his 2014 book–says about Mach's *new* realism: Mach's "fundamental views are essentially different from those of idealistic philosophers […]. We may even call them realistic, but their realism is essentially different from so-called philosophical realism" (Kleinpeter 1906, 163-164).

We might be perspectivists to the extent that we endorse both (1) and (2), but accepting either one of them is sufficient to reject the radical form of metaphysical realism that implies that "the world consists of some fixed totality of mind-independent objects" and that "there is exactly one true and complete description of 'the way the world is'" (Putnam 1981, 49; quoted in Massimi 2018a, 165). "As soon as we acknowledge that our scientific knowledge is always from a specific vantage point",[27] which might be interpreted *diachronically* (i.e. depending on the historical component) or *synchronically* (i.e. focused on the alternative research programmes or modelling practices developed in different cultural frameworks but at the same time), we might realize that "there cannot be an objective, unique, true description" of the state of affairs (Massimi 2018a, 165). But if this is the case, if we maintain that our world-description is influenced by our contingent viewpoint, that "no knowledge of nature is possible outside the boundaries of historically well-defined scientific perspectives", but also that "the representational models are themselves perspectival in idealizing some factors and abstracting from another" (2018a, 166-167), then "the realist's *truth* and *objectivity* must be called into question", and we must deal with the problematic issue of relativism (2018a, 168).

The originality of Massimi's (as well as Giere's) defence of perspectivism resides precisely in her attempt to show that this view is in fact compatible with realism–although interpreted in a moderate way, incorporating constructivist features. Given that it is possible to have a plurality of models that abstract from the same target system, and given that we might refer to this system in order to assess these models, it seems to be possible not to abandon the realist tenet once and for all. In other words, we should explore the possibility of defending realism *within synchronic perspectives*.

Massimi indeed endorses *epistemic pluralism* as a rejection of objectivist realism that does not necessarily speak against perspective-independent facts. For her, "one can accept and fully endorse that scientific inquiry is indeed pluralistic and that there is no unique, objective, and privileged epistemic vantage point without necessarily having to conclude that perspectives shape scientific facts or relativize truth" (Massimi 2018a, 170). Based on this, Massimi provides us with a new, viable account of perspectival realism:

> Philosophers on either side of the debate on perspectivism have at best failed to draw a clear-cut distinction between *objectivity* and *truth*; at worst, they have conflated the two. For often enough perspectivism is presented as a view about *facts* being perspectival; or about *properties* being relational; or about *truth* being relative. Couched in this language, it is no surprise that perspectivism verges on either fact-constructivism or alethic relativism–and, needless to say, either option is a non-starter for cashing out a viable form of perspectival *realism* (if the title *realism* is still meaningful). If perspectivism has to be made compatible with realism, something ought to be said about facts not being shaped by scientific perspectives or truth relativized to them. (Massimi 2018a, 170)

[27] Massimi traces our acknowledgement of the human vantage point back to Kant's Copernican Revolution (Massimi 2018a, 166). Quite interestingly, she remarks that "Kant's *perspectival knowledge* is […] one of the greatest legacies of the philosophy of the Enlightenment: it gives the best answer to the question 'what is scientific knowledge?', by providing an antidote […] against the ongoing dangers of bogus knowledge and popular opinions" (Massimi 2017, 81; cf. also 65).

For Massimi, there is no direct and necessary relationship between the rejection of scientific *objectivity* and the idea that the worldly states of affairs is *relative* to scientific perspectives. On the contrary, perspectival realism "can share with scientific realism the view that worldly states of affairs, the language of science, and truth as correspondence are all perspective independent. What makes [it] perspectival [is that] the rejection of the God's eye view leads to a genuinely novel and fruitful notion of *perspectival truth* and *scientific progress across perspectives* (among several other relevant aspects)" (Massimi 2018a, 171). The goal is therefore to provide a revaluation of scientific knowledge–a reassessment of its value and meaning that makes sense of concepts such as truth and objectivity from a different viewpoint, without necessarily getting rid of them once we acknowledge their perspectival character. Put differently, we must critically enlighten scientific knowledge in order to properly assess its contingent character, in a way that allows us to save it from sceptical relativist *non sequiturs*. But this also involves a redetermination of what "realism" might mean within the perspectival framework, as Massimi correctly observes, and this is a crucial point that we should not neglect when dealing with perspectival realism. What is needed, in order to do that, is a clear explanation of "what it is like to be *true within a perspective*", of what a "perspectival truth" conceived of as "truth […] *contextualized* within the limits afforded by rival scientific models or rival historical perspectives" might actually be (Massimi 2018a, 171).

In dealing with these issues, Massimi defends "the metaphysical tenet that states of affairs about the world are *perspective-independent*; whereas our scientific knowledge claims about these states of affairs are *perspective-dependent*" (2018b, 342), and further argues that the notion of perspectival truth should provide us with a link between the two planes, showing that our scientific knowledge claims can be perspectival while also being claims about the world as *it is*. In other words, perspectival truth might allow us to argue that "it is possible to be realist about science, while also taking on board the situated and perspectival nature of scientific knowledge" (2018b, 343). Interestingly, Massimi argues that perspectives provide "contextual truth-condition[s]" which might permit inter-individual, shared views (2018a, 171), and she focuses on *perspectival standards of performance adequacy* as prerequisites that must be satisfied if scientific knowledge is to be properly assessed. As she explains, this expression "indicate[s] the standards that must be met by scientific knowledge claims for them to be retained across scientific perspectives, i.e. for their ongoing performance to be judged as adequate by practitioners of different scientific perspectives. Hence, *standards of performance-adequacy* capture the properly contextual, perspectival, and pragmatic features of the notion of truth within perspectives" that Massimi tries to cash out (2018b, 354 fn.).[28]

This conception is consistent with Massimi's view that scientific progress may actually track worldly states of affairs across scientific perspectives and that these states of affairs must ultimately be "the tribunal that decides whether any knowledge claim is true or false" (Massimi 2018a, 172), but also that "correspondence with the world" is barely enough, for the situated character of scientific knowledge also plays an essential part in the whole picture and cannot be discarded. But it is worth remarking that, for Massimi, perspectival

[28] Cf. also Massimi 2018b, 355: "*Accuracy* with respect to fundamental mathematical equations; *empirical testability* within the limits of well-defined tests; *projectibility* and *heuristic fruitfulness* across a variety of engineering practices, are just a few examples of the kind of standards of performance-adequacy that perspectival contexts of use may put in place for determining the truth-conditions of scientific knowledge claims".

realism only deals with the *epistemological* tenet of scientific realism, i.e. it only calls into question the actual value of our scientific knowledge claims, while admitting the existence of perspectival-independent states of affairs. In fact, she argues that "the standards of performance adequacy do not fulfil the task of warranting what we can assert about the electron or the properties of water; nor do they claim to replace truth as correspondence. Instead, their task is simply to allow us to evaluate the ongoing performance of our scientific knowledge claims across time and perspectival shifts, because we simply do not possess a God's eye view to do that otherwise" (Massimi 2018b, 358).

Perspectival realism thus seems to be capable of merging realism with relativism, in an account that makes sense of the constructivist and instrumental side of scientific knowledge while demanding the latter's testability on the empirical plane of facts. It does not matter how extensively situated a theory might be; it should keep pace with facts anyway, and our assessment of it must take into account both the theoretical and the empirical plane. This can be the starting point for a tentative reflection on the consistency between Mach's *Erkenntnislehre* and perspectival realism. Given that these views are the expression of different strategies of inquiry, with a different aim and scope–for they are as historically and culturally situated as their subject matter–it seems to be possible to compare some of their features, starting from the way Mach calls into question the *value* of scientific knowledge by stressing the "mental economical", i.e. abstract, character of its claims. As I have tried to show above, his reflection on the relationship between theories and facts, and his conclusion that science is situated both historically and within the perspective of the individual inquirer, allowed Mach to defend a relativization of the truth value of scientific knowledge claims, while admitting pragmatist standards of assessment that take into account the response that our theories receive from the facts they aim to explain, thus permitting Mach to preserve an ideal of scientific progress. Furthermore, Mach seems to be at least mildly inclined to admit that science aims at providing us with "as faithful a picture as possible of the course of nature herself" (Mach 1976, 81–82), a statement which might be compared with Massimi's view of science as tracking perspective-independent states of affairs. Although I believe that Mach would have been quite sceptical about the metaphysical tenet endorsed by Massimi, and although the extent to which the factual basis of which Mach speaks is real or not is one of the most problematic issues of his epistemology (a problem that can perhaps be solved only through the redefinition of the notion of "real" itself), the 1910 paper gives us something to reflect upon. Indeed, in that paper, Mach argues that "the *human* world picture which is *socially* preserved by scientists succeeding or replacing each other *will be more independent of individuality, progressively giving a purer expression of the facts*" (Mach 1910, 230, emphasis added). That is, the development of scientific knowledge may allow us to refine our perspectival view and thus to elaborate a more "objective" one–a view which may be independent of individual viewpoints. Of course, this does not mean that science actually *tracks* perspective-independent states of affairs, for Mach, but rather that it can at least be viewed as the expected ideal outcome of a process of adaptation of thoughts to facts, as Mach conceived of it. That is, we should contextualize the observation published in the 1910 paper within a moderate realist picture that admits that the direct mirroring of facts in theories is a mere ideal that may never be achieved (cf. e.g. Mach 1892, 202; 1895, 187; 1986, 361; 1976, 355) while considering that the value of scientific knowledge claims is not only a matter of coherence, for Mach, but depends on the pragmatic adequacy of theories in relation to observations.

The pragmatist features of Mach's epistemology can actually be helpful for a further (and final) consideration. The relationship between pragmatism and perspectivism has recently been explored by Hasok Chang. For him, these views "share a deep humanist impulse, which is to regard science as a thoroughly human activity, even when it is aimed at the production of the most abstract and objective kind of knowledge" (Chang 2019, 10). Like perspectivism, pragmatism takes into account the situated character of human knowledge and tries to keep pace with our actual experience, with no need for a foundational criterion of assessment. But this is precisely the point: given that we observe the world only from the human viewpoint, and given that even if we take experience to be the ultimate source of learning, we are still building our world-description on a contingent relationship with the states of affairs, we must admit that "there cannot be any knowledge that is not perspectival" (Chang 2019, 20). This leads Chang to a stronger kind of perspectivism that may be closer to Mach's view than Massimi's. "All we can ever talk about", he argues, "are conceptualized objects, which are in the realm of Kantian phenomena rather than things-in-themselves". Furthermore, "one can argue that the relation between our knowledge and the world cannot be spelled out in a straightforward way", i.e. we cannot "take it for granted that the three-dimensional objects we are perspectivally studying exist 'out there' in themselves, well-formed independently of all our cognition and action" (Chang 2019, 20–21). With this, Chang does not endorse a strong ontological perspectivism, admitting that "there are no perspective-transcendent ontological facts or states of affairs" (Chakravartty 2017, 177; quoted in Chang 2019, 22). Rather, he only tries to moderate the realist tenet that the scientific picture of the world corresponds to a ready-made reality, which Massimi also endorses, if my reading is correct. For Chang, "any phenomenon of nature that we can think or talk about at all is couched in concepts, and we choose from different conceptual frameworks, which are liable to be incommensurable with each other" (Chang 2019, 22).

As noted above, I believe that this version of perspectivism may be closer to Mach's epistemology–or conversely, that some of the main features of Mach's conception of scientific knowledge may support Chang's view. They both agree that science is a human, historically situated enterprise, that experience is the ultimate source of learning, that our world-description is couched in thoughts, and that we can speak only about conceptualized objects or facts–which are themselves perspectival, in more than one sense. When combined, these tenets outline a moderate constructivist or modest realist and pragmatically oriented view of scientific knowledge, a conception that argues that the value of true scientific statements must be relative while maintaining that a criterion of objectivity can still be found by reflecting on the relationship between theories and facts which takes place within the system of knowledge claims provided by scientific inquiry.

3 Conclusion

Mach contributed significantly to the development of the contemporary philosophy of science. His reception of the philosophical approach that was standard for his time and his attempt to further elaborate it in an original way represented an important step toward developing attitudes that are currently endorsed in the discipline. The enlightenment side of his method of inquiry aimed to call into question the value of scientific knowledge claims by stressing their being *situated* historically, culturally, and within an individual viewpoint. In fact, historical analysis–the fundamental tool of Mach's epistemology–gives prominence to this situatedness, but it also presupposes the epistemological tenet that science is a mental-economical enterprise that delivers conceptual *abstractions*, i.e. *indirect descriptions*. Yet a positivistic ideal of scientific progress seems to be preserved in the view that ideas may *adapt* to facts adequately, at some stage–that science aims to *faithfully* mirror facts in thought. This conception therefore outlines a multifaceted picture of scientific knowledge that is indeed an *instrumental* picture, one that does not give up completely on a *moderate realist* commitment that conceives of facts as the reference for a *pragmatic* assessment of our knowledge claims.

As I have tried to argue, these features of Mach's epistemology might dialogue profitably with the *perspectival* view that has been defended recently in the philosophy of science. The situated character of scientific knowledge, the idea that the *goal* of scientific inquiry is to track states of affairs about the world that are *perspective-independent*, even though our scientific knowledge claims about these states of affairs are *perspective-dependent*, and the broader and fundamentally pragmatist tenet that science is a thoroughly human activity and that any phenomenon of nature that we can think of or talk about at all is couched in concepts–all of these elements can be found in Mach (although the general framework is different, i.e. each view must be contextualized within a specific historical and cultural perspective). Thus, although it would be inaccurate to argue that Mach was a perspectival realist himself, it seems possible to say that some of the ideas that follow from his original methodology for a critical (philosophical) analysis of science cleared the way for further reflection on the value of scientific knowledge claims, one of the most recent and promising outcomes of which is perspectival realism.

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References

Banks, E. C. (2003). *Ernst Mach's World Elements. A Study in Natural Philosophy*. Kluwer.
Banks, E. C. (2004). The Philosophical Roots of Ernst Mach's Economy of Thought. *Synthese* 139: 23–53.
Banks, E. C. (2014). *The Realistic Empiricism of Mach, James, and Russell: Neutral Monism Reconceived*. Cambridge University Press.
Banks, E. (2021). The Case for Mach's Neutral Monism. In *Interpreting Mach. Critical Essays*, ed. J. Preston, 258–279. Cambridge University Press.
Blackmore, J. T., ed. (1972). *Ernst Mach: His Work, Life, and Influence*. University of California Press.
Blackmore, J. T., ed. (1992). *Ernst Mach – A Deeper Look*. Springer.
Bradie, M. (1986). Assessing Evolutionary Epistemology. *Biology and Philosophy* 1: 401–459.
Chakravartty, A. (2017). *Scientific Ontology*. Oxford University Press.
Chang, H. (2019). Pragmatism, Perspectivism, and the Historicity of Science. In *Understanding Perspectivism. Scientific Challenges and Methodological Prospects*, eds. M. Massimi, and C. D. McCoy, 10–27. Routledge.
Čapek, M. (1968). Ernst Mach's Biological Theory of Knowledge. *Synthese* 18: 171–191.

De Waal, E., and S. L. Ten Hagen. (2020). The Concept of Fact in German Physics around 1900: A Comparison between Mach and Einstein. *Physics in Perspective* 22: 55–80. https://doi.org/10.1007/s00016-020-00256-y.
Feyerabend, P. (1970). Philosophy of Science: A Subject with a Great Past. In *Historical and Philosophical Perspectives of Science*, ed. R. Stuewer, 172–183. University of Minnesota Press.
Feyerabend, P. (1984). Mach's Theory of Research and its Relation to Einstein. *Studies in History and Philosophy of Science* 15: 1–22.
Frank, P. (1970). The Importance of Ernst Mach's Philosophy of Science for Our Times. In *Ernst Mach: Physicist and Philosopher*, eds. R. S. Cohen, and R. J. Seeger, 219–235. Reidel.
Giere, R. (2006). *Scientific Perspectivism*. University of Chicago Press.
Goeres, R. (2004). Sensualistischer Phänomenalismus und Denkökonomie. Zur Wissenschaftskonzepzion Ernst Machs. *Journal of General Philosophy of Science* 35: 41–70.
Gori, P. (2018). Ernst Mach and Pragmatic Realism. *Revista Portuguesa de Filosofia 74*(1): 151–172.
Gori, P. (2021). Ernst Mach's Contribution to the Philosophy of Science in Light of Mary B. Hesse's Postempiricism. *HOPOS – The Journal of the International Society for the History of the Philosophy of Science 11*(2): 383–411.
Guzzardi, L. (2021). Holding the Hand of History. Mach on the History of Science, the Analysis of Sensations, and the Economy of Thought. In *Interpreting Mach. Critical Essays*, ed. J. Preston, 164–183. Cambridge University Press.
Haller, R. (1988). Grundzüge der Machschen Philosophie. In *Ernst Mach. Werk und Wirkung*, eds. R. Haller, and F. Stadtler, 64–86. Hölder-Pichler-Tempsky.
Hesse, M. (1980). *Revolutions and Reconstructions in the Philosophy of Science*. Harvester Press, and Indiana University Press.
Hiebert, E. (1970). Mach's Philosophical Use of the History of Science. *Minnesota Studies in the Philosophy of Science* 5: 184–203.
Holton, G. (1988). *Thematic Origins of Scientific Thought* (rev. edition). Harvard University Press.
Hui, A. (2013). Ernst Mach's and Max Plank's Studies of Accomodation in Hearing. *Osiris* 28(1): 119–145.
Hui, A. (2021). Mach's Piano and the Making of a Psychophysical Imaginarium. In *Interpreting Mach. Critical Essays*, ed. J. Preston, 10–27. Cambridge University Press.
Kant, I. (2004). *Prolegomena to Any Future Metaphysics That Will Be Able to Come Forward as Science*. Eng. transl. by G. Hatfield. Cambridge University Press.
Kleinpeter, H. (1906). On the Monism of Professor Mach. *The Monist 2*(2): 198–168.
Mach, E. (1892). Facts and Mental Symbols. *The Monist 2*(2): 198–208.
Mach, E. (1894). On the Principle of the Conservation of Energy. *The Monist 5*(1): 22–54.
Mach, E. (1895). *Popular Scientific Lectures*. Open Court.
Mach, E. (1910). Die Leitgedanken meiner naturwissenschaftlichen Erkenntnislehre und ihre Aufnahme durch die Zeitgenossen, *Scientia* 4, 225–240. [Part of the paper is published in English translation in Blackmore, J., ed. 1992. *Ernst Mach – A Deeper Look*, 133–139. Springer.].
Mach, E. (1911). *History and Root of the Principle of the Conservation of Energy*. Open Court.
Mach, E. (1919). *The Science of Mechanics: A Critical and Historical Account of its Development*. Open Court.
Mach, E. (1959). *The Analysis of Sensations*. Dover.
Mach, E. (1976). *Knowledge and Error. Sketches on the Psychology of Enquiry*. Springer.
Mach, E. (1985). *Die Analyse der Empfindungen und das Verhäaltnis des Physischen zum Psychischen*. WBG.
Mach, Ernst. (1986). *Principles of the Theory of Heat*. Reidel.
Mach, E. (2008). *Die Analyse der Empfindungen und das Verhältnis des Pysischen zum Psychische*. Xenomoi.
Mach, E. (2016). *Die Prinzipien der Wärmelehre: historisch-kritisch entwickelt*. Xenomoi.
Massimi, M. (2017). What is this Thing Called 'Scientific Knowledge'? – Kant on Imaginary Standpoints and the Regulative Role of Reason. *Kant Yearbook 9*(1): 63–83.
Massimi, M. (2018a). Perspectivism. In *The Routledge Handbook of Scientific Realism*, ed. J. Saatzi, 164–175. Routledge.
Massimi, M. (2018b). Four Kinds of Perspectival Truth. *Philosophy and Phenomenological Research 96*(2): 342–359.
Nemeth, E. (2019). Zur "historisch-kritischen Methode" bei Ernst Mach. In *Ernst Mach – Zu Leben, Werk und Wirkung*, ed. F. Stadler, 21–43. Springer.
Patton, L. (2019). New Water in Old Buckets: Hypothetical and Conterfactual Reasoning in Mach's Economy of Science. In *Ernst Mach – Life, Work, Influence*, ed. F. Stadler, 333–348. Springer.
Patton, L. (2021). Abstraction, Pragmatism, and History in Mach's Economy of Science. In *Interpreting Mach. Critical Essays*, ed. J. Preston, 142–163. Cambridge University Press.

Pihlström, S. (2007). Metaphysics with a human face: William James and the prospects of pragmatist metaphysics. *William James Studies* 2(1): 1–28. http://williamjamesstudies.org/metaphysics-with-a-human-face-william-james-and-theprospects-of-pragmatist-metaphysics/.
Pihlström, S. (2009). *Pragmatist Metaphysics*. Continuum.
Pihlström, S., N. Weidtmann, and F. Stadler, eds. (2017). *Logical Empiricism and Pragmatism*. Springer.
Pojman, P. (2011). The Influence of Biology and Psychology upon Physics: Ernst Mach Revisited. *Perspectives on Science 19*(2): 121–135.
Preston, J. (forth.). Paul Feyerabend's Ernst Mach (in preparation).
Putnam, H. (1981). *Reason, Truth, and History*. Cambridge University Press.
Putnam, H. (1995). *Pragmatism. An Open Question*. Blackwell.
Sass, H. M. (1976). Immanenzphilosophie. In J. Ritter and K. Gründer, eds. *Historisches Wörterbuch der Philosophie*, vol. 4, 238–240. Schwabe.
Staley, R. (2017). "Beyond the Conventional Boundaries of Physics": On Relating Ernst Mach's Philosophy to His Teaching and Research in the 1870s and 1880s. In *Integrated History and Philosophy of Science*, ed. F. Stadler, 69–80. Springer.
Staley, R. (2018). Sensory Studies, or When Physics was *Psychophysics*: Ernst Mach and Physics Between Physiology and Psychology, 1860–71. *History of Science 59*(1). https://doi.org/10.1177/0073275318784104.
Uebel, T. (2019). Mach, Jerusalem and Pragmatism. In *Ernst Mach – Life, Work, Influence*, ed. F. Stadler, 501–523. Springer.
Uebel, T. (2021). Ernst Mach's Enlightenment Pragmatism. History and Economy in Scientific Cognition. In *Interpreting Mach. Critical Essays*, ed. J. Preston, 84–102. Cambridge University Press.
von Mises, R. (1987). Ernst Mach and the Scientific Conception of the World. In *Unified Science*, ed. B. McGuinness, 166–190. Reidel.
Williams, B. (1985). *Ethics and the Limits of Philosophy*. Harvard University Press.